The Skipped Candidates in the NIST and ANSI X9.62 Curve Seed Runs: A Negative Result, and a Structural Observation on the K-163 Generator

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Abstract

In October 2026 Anthropic showed that the SEED values in FIPS 186-2 and ANSI X9.62 are not hashes of anything: each is 20 bytes of C-library `lrand48` output seeded by a Unix timestamp, and a published curve is candidate number n in a run that drew candidates in sequence. The same reproduction exposes a loose end: in the four prime-field runs, eleven earlier candidates also give prime-order curves of the right size that pass every documented criterion, yet eight later ones were published. We ask what rejected the eleven. We regenerate all four runs, verify every group order, and test roughly fifty invariants of the nineteen viable curves across fifteen families—the arithmetic of n and t, the CM discriminant and class number, isogeny structure over $\mathbb{F}_p$ and over extension fields, extension-field group orders, embedding degree, gcds of $p \pm 1$ with $n \pm 1$, Legendre symbols and residue classes, and the small-x point structure—together with a dozen models of the generating program, including eleven simulated early-abort bugs and a set-cover search for unions of small defects. Nothing separates the eleven skipped curves from the eight published ones; the skipped set is no easier to cover by "bug-shaped" predicates than a random relabelling. We conclude the skips are unexplained by any property of the curves and most plausibly reflect run-time state or a human choice that left no numerical trace. Two facts about the program do emerge: it computed b with the $(p+1)/4$ square root for $p \equiv 3 \pmod 4$, and it drew exactly 20 bytes per candidate. Separately, extending a small-multiple scan to the binary curves, we find that the half of the published K-163 generator has a 126-bit x-coordinate on a 163-bit curve—a 2^{-37} coincidence, or the signature of a 128-bit chosen value doubled into the prime-order subgroup. We do not identify the value; we report it as evidence that the base points, unlike the seeds, still have structure to be recovered.

1 Introduction

ECDSA's standard curves were handed to NIST in 1997–98 with seed values and no account of their origin [7, 8]. Koblitz and Menezes catalogued the resulting unease [5]; Bernstein, Lange and coauthors showed how much freedom a seed-search would have given a malicious designer [6]; Valsorda posted a bounty for the seeds' derivation [4]. In October 2026 Anthropic's Frontier Red Team reported that the seeds are `lrand48` output: five NIST prime-curve seeds and five ANSI X9.62 example-curve seeds are 20 consecutive bytes of `lrand48`() $\gg 23$ after `srand48`(t) with t a Unix timestamp between May 1997 and April 1998, and four of the five binary-curve seeds are

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five 31-bit `lrand48` words from one sitting on 17 April 1998 [1]. Valsorda reproduced the result and claimed the bounty [3]. The work is one output of a broader programme of model-assisted cryptanalysis at Anthropic [2], whose authors identify human validation of model-found results as the current bottleneck; the present paper is, among other things, such a validation. The base points of P-192 and the ANSI example curves were also explained, as output of FreeLIP's `zrandom` with small integer seeds.

Both write-ups note a residue. A run produces candidates in order; a candidate is *valid* if $-27/c$ is a square mod p (so a coefficient b exists), *viable* if the resulting curve has prime order n with $\lceil \log_2 n \rceil \leq \lceil \log_2 p \rceil$ (an undocumented but evidently applied rule [3]). In the P-256 run, candidate 876 is viable but candidate 1842 was published; the P-224 run passed over three viable candidates and the two ANSI runs seven more. Valsorda summarizes: "roughly twice as many valid seeds as were published," with a second undocumented criterion that "still escapes us."

This paper is the attempt to find that criterion. It fails, and the failure is the result: the eleven skipped curves are, on every cheaply computable property, indistinguishable from the eight published ones. Section 2 fixes the data. Section 3 reports the invariant sweep, Section 4 the program models, Section 5 the set-cover test. Section 6 reports the K-163 observation. Section 7 lists what remains open.

2 Setup and reproduction

Runs. The four prime-field runs and their published candidates are: P-256 (`srand48` 1997-08-16 01:52:34 UTC, candidate 1842); P-224 (1998-04-03 12:52:07, candidate 7026); the prime192 run (1997-05-18 14:45:34; P-192 at 139, prime192v3 at 5228, prime192v2 at 6013); the prime239 run (1997-05-15 15:49:03; prime239v1/v2/v3 at 2185, 2320, 3048).

Reproduction. For every candidate we derive c by the X9.62 SHA-1 expansion and both square roots of $-27/c$. Candidates with 2-torsion (the cubic has a root) are discarded without counting. Group orders are computed with full SEA in PARI/GP 2.10 built from source and verified by $n \cdot P = O$ on random points, which for prime n in the Hasse interval determines $\#E$. (An early-abort pass produced trace-sign errors in this build and was discarded.)

Square-root convention. For every published curve over $p \equiv 3 \pmod 4$—P-256, P-192, prime192v2, prime192v3, prime239v1, v2, v3—the published b equals $(-27/c)^{(p+1)/4} \bmod p$, never the other root. The program computed that root and tested only that curve. The other root is its quadratic twist, prime for $4 + 10 + 9$ further candidates across the three runs which the program never saw; a reproduction accepting either root overcounts badly. For P-224 ($p \equiv 1 \pmod 4$) the roots give isomorphic curves.

The nineteen. With that convention and the size rule, the viable candidates are

run	viable	published	skipped
P-256	876, 1842	1842	876
P-224	777, 1485, 5736, 7026	7026	777, 1485, 5736
prime192	139, 2198, 5228, 5405, 6013	139, 5228, 6013	2198, 5405
prime239	36, 1562, 2185, 2320, 2491, 2682, 2863, 3048	2185, 2320, 3048	36, 1562, 2491, 2682, 2863

eight published, eleven skipped—exactly the lists in [1, 3]. In every run the published candidate with the largest index is the last viable candidate of the run; this is where runs stopped, not a selection rule, since 139 and 5228 (and 2185, 2320) were published with later viable candidates skipped.

Calibration. The unknown criterion rejected about half of the viable candidates, not one in a thousand of all candidates. Any proposed rule must fail at that rate.

3 Curve invariants

Every property below was computed for all nineteen curves and compared between the published and skipped groups. The strongest form of the negative evidence is in the two runs that published three curves each: there the published curves disagree *with one another* on most binary properties, so no such property can have been required. Tables are in Appendix A; the full record with scripts is in the repository.

3.1 Arithmetic of n and t

Twist order $n' = 2p+2-n$: small-factor structure and cofactor primality interleave (published P-224: $3^2 \cdot 11 \cdot 47\times$ 212-bit composite; skipped 1485: $3 \cdot 31\times$ 218-bit composite). Complete factorizations of $n \pm 1$ (PARI MPQS; all cofactors fell in seconds): the published P-256 has the smoothest $n-1$ of its run (largest prime factor 92 bits) while the candidate it displaced has $n - 1 = 2 \cdot 3 \cdot 7 \cdot 11 \cdot 43 \times P_{242}$; in prime239 the published largest prime factors (69, 128, 110 bits) sit inside the skipped range (76–219). Congruences of t, n, b modulo 2, 3, 4, 5, 8, 16; $b < p/2$; $(n-1)/2$ prime: nothing.

3.2 CM structure

With $D = t^2 - 4p = fs^2$, f fundamental, the conductor s of $\mathbb{Z}[\pi]$ in $\mathcal{O}_K$ is 1 for sixteen curves, 3 for the published P-224, and 5 for one published (prime239v2) and one skipped (2682) prime239 curve. The class number, estimated as $\sqrt{|f|}\, L(1, \chi_f)/\pi$ with the Euler product to 10^6, is *smallest* in its run for the published P-224 (by a factor ≈ 4.5, from both the 3^2 and a small L-value) and *larger* for the published P-256 than for its skipped competitor; in the ANSI runs the values interleave. Neither a floor nor a ceiling on class number is consistent with the data. We note that the heavy tail of the largest-square-divisor statistic—$\Pr(s \geq X) \approx 1/X$, not $1/\sqrt{p}$—made a class-number floor *statistically admissible* as a criterion rejecting half the candidates; the data simply say it was not applied.

3.3 Isogeny structure

For $s = 1$, $\mathrm{End}(E) = \mathcal{O}_K$ is forced; the only structure is the splitting of small ℓ in K, read off as the number of roots of $\Phi_\ell(j(E), X)$ mod p for $\ell \leq 43$. The patterns are random: 2 inert everywhere (forced: n odd), $3 \mid D$ for all six P-256/P-224 curves but inert in both ANSI runs ($p \equiv 2 \pmod 3$), 11 split for the six P-256/P-224 curves (a 1.5% coincidence not shared by the wider candidate set). The published P-224 ($s = 3$) has $\Phi_3(j, X)$ with four roots: the crater of a height-one 3-volcano. It is the only curve of the six with any volcano structure, and it is published. Over $\mathbb{F}_{p^k}$ the Frobenius conductor is $s\,U_k$ with U_k the Lucas sequence; volcano depths $v_\ell(sU_k)$ for $\ell \leq 47$, $k \in \{2, 3, 4, 6, 12\}$ vary widely (depth 7 at $\ell = 2$, $k = 3$ for skipped 777; depth 5 at $\ell = 5$, $k = 3$ for published 2185) and do not sort.

3.4 Extension-field group orders

$\#E(\mathbb{F}_{p^k})/n$ for $k \le 12$, trial-divided to 10^6: smooth parts of 1–150 bits, as for random integers of these sizes, interleaving at every k.

3.5 Embedding degree, gcds, symbols

$\mathrm{ord}_n(p) \ge 2^{222}$ or $> 2 \cdot 10^4$ for every curve. $\gcd(p \pm 1, n \pm 1)$ and $\gcd(p^2 - 1, n^2 - 1)$ are small products of small primes; no gcd statistic (value, odd part, largest prime factor) ranks published above or below skipped consistently across the four runs—$\gcd(p^2 - 1, n^2 - 1)$ puts the published *below* the skipped in P-256 and *above* in P-224 and prime192. Legendre symbols $(n/p), (p/n), (t/p), (t/n), (b/p), (-3/n), (-1/n), (2/n)$ and residues of n, p, t mod 4 and 8: within prime192 the published curves have $(n/p) = +1, -1, +1$ and $n \equiv 1, 3, 1 \pmod 8$; within prime239, $n \equiv 3, 3, 1 \pmod 8$. No symbol or residue is constant on the published curves across runs.

3.6 Point structure

The smallest $x \ge 0$ with a point is 0 for sixteen curves (iff b is a square), 1 for two skipped P-224 curves, 3 for the published P-224. The y at that x, taken with the $(p + 1)/4$ root, is full-size everywhere. On-curve bitmaps for $x \in [0, 255]$ are 256 fair coin flips per curve (118–148 on-curve; published mean 128.2, skipped 129.6; the one 2.5σ value is published prime239v1). No small-x point is a scalar multiple ($k \le 2^{20}$) of another or of the published generator.

4 Models of the program

RNG stride. The published indices within a run are coprime (139, 5228, 6013; 2185, 2320, 3048), and each published seed sits at draw offset exactly $20(i - 1)$. Any extra RNG consumption therefore came in whole 20-byte blocks. Rules of the form "after a candidate with property X, skip k blocks" for six properties X (valid b; even order; odd order; odd with no small factor; prime but too large; prime on the twist root), singly and in pairs, $k \le 7$: none visits all eight published seeds while missing all eleven skipped; the best scores 13 of 19 against a random-subset expectation of 9.5.

Simulated early-abort bugs. Since t and n are known exactly, any rule "reject if statistic $\equiv 0 \pmod \ell$ for some $\ell \le 100$" is simulable. Eleven were tried—the correct abort (control: rejects nothing), trace sign flipped, order of the other root, both flipped, $t \equiv 0$, $t \equiv \pm 1, \pm 2$, off-by-one on n in both directions, signed/unsigned confusion, and a twist-order abort—and none separates (Appendix A). Bounded versions (first $\ell \le B$, $B \le 1000$) and single-residue rules $X \equiv r \pmod \ell$ for nine statistics likewise fail. A bug that misbehaved *consistently* is excluded; one that depended on run-time state is not recoverable from the parameters.

Time limit. Elkies/Atkin balance for $\ell \le 97$: published curves average 9.9 Elkies primes of 24 against 12.5 for skipped. A timeout would have cut the slow (Atkin-heavy) curves, i.e. the published ones.

Index-dependent bugs, seed-byte mangling, base-point step, human sorting. The eleven skipped indices share no residue class, bit, popcount or buffer-offset property disjoint from the eight published. Two skipped seeds (and no published seed) contain a 0x00 byte—consistent with a C-string bug for those two and no others; 0x0A appears on both sides; a bit-length check is excluded

by the published P-192 seed (158-bit value). The FreeLIP base-point recipe, run on every viable prime192/prime239 curve with the published seeds and 600 nearby ones, behaves identically on published and skipped curves (control: the six published seeds reproduce the six published G_x). In the three-curve runs the published curves are not the top-3 or bottom-3 under index, seed, b, n, t or $n - p$.

5 A union of small bugs? Set cover

Each test above asks for a single rule. A union of two or three unrelated defects would also explain why the published curves disagree among themselves. From 135 cheap binary predicates on seed bytes, n, t, b, index, small residues and symbols, 29 are false on all eight published curves; the smallest union of them covering all eleven skipped has size 4 (ten such covers, all residue-class predicates such as $20(i-1) \bmod 4096 < 1024 \cup b \equiv 1\ (7) \cup n \equiv 7\ (8) \cup t \equiv 0\ (7)$). On 300 random relabellings preserving per-run counts, a cover of size ≤ 4 exists 35% of the time. The true labelling is unremarkable; the covers it admits are noise fitted to noise.

6 The K-163 generator

The base points of P-224, P-256, P-384 and P-521 remain unexplained [1]. A folk hypothesis—that a generator was produced by hashing a phrase to an x-coordinate and doubling—predicts a short coordinate at $G/2$. For the ten prime-field generators, kG and G/k for $2 \leq k \leq 2^{16}$ (negations included) have no coordinate shorter than the field size minus ≈ 20 bits, exactly the minimum expected from 2^{17} random elements.

Over $\mathbb{F}_{2^m}$ the same scan on the ten NIST binary curves (kG, $k \leq 2^{16}$; G/k, $k \leq 2^{12}$; parameters verified by $G \in E$ and $nG = O$) finds nine random-looking curves and one that is not. On K-163 ($y^2 + xy = x^3 + x^2 + 1$, cofactor $h = 2$),

$$G/2 = \tfrac{n+1}{2}\,G \quad \text{has} \quad x = \texttt{0x3a5da90060df8af5be2121e601ae11f1},$$

a 126-bit value—128 bits with two leading zeros—while y is 161 bits. A coordinate 37 bits short is a 2^{-37} event; across the $\approx 7 \times 10^4$ points scanned the expected number of such hits is 10^{-6}. The natural reading: a Koblitz curve has no seed ($b = 1$), so G is its only chosen parameter; a 128-bit x was chosen, y solved, and the point multiplied by the cofactor into the prime-order subgroup. The other Koblitz curves ($h = 4$) show nothing at $G/4$, nor do the B curves at $G/2$. The value is not the MD5 or SHA-1 of a handful of obvious strings and is not four `lrand48` words.

We have not identified the value, and we make no claim about what it is. We include the observation because of what it shows about the state of the problem. Anthropic's result settled the seeds completely; it is tempting to read that as closing the provenance question for the NIST curves. The K-163 half-point says otherwise: at least one published generator carries a structural fingerprint of its construction that nobody has decoded, and the method it suggests (choose x, solve y, multiply by the cofactor) is not the FreeLIP method that produced the May 1997 base points. The base points are a second, separate provenance problem, with at least one concrete handle on it.

7 Open problems

1. The eleven skips: by the results above, only a run-time-dependent bug or a non-numerical human choice remain. Archival evidence (the program, or a printout of accepted curves) would settle it; nothing in the parameters will.

2. The 128-bit value behind K-163's $G/2$: hash preimage search over 1997-era phrases, encodings and hash functions; or an RNG stream. A match would date and explain the binary-curve base points as the `lrand48` result did the seeds.

3. Base points of P-224, P-256, P-384, P-521, the five B curves and the four other K curves. If the K-163 method (choose x, solve y, multiply by h) was general, x was full-size on the others—a hash expanded to field size, or RNG output, possibly from the same 17 April 1998 session as the B seeds.

4. The B-163 seed, which fits no RNG tried in [1].

Reproducibility

Scripts, verified orders for all candidates in the four runs, and raw outputs are in the accompanying repository. Point counting uses PARI/GP 2.10 built from `https://github.com/jpflori/pari` (no `seadata`; modular polynomials computed on the fly). Binary-field arithmetic uses PARI's `ffgen`. All group orders carry an independent $nP = O$ check.

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A Tables

A.1 Class number and $n \pm 1$ for all viable curves

Generic $\log_2 h \approx \log_2 \sqrt{4p}$: 129.0 (P-256), 113.0 (P-224), 97.0 (prime192), 120.5 (prime239). s is the conductor of $\mathbb{Z}[\pi]$; L is $L(1, \chi_f)$ by the Euler product to 10^6; LPF = largest prime factor, in bits.

run	cand	s	$L(1,\chi)$	$\log_2 h$	LPF(n−1)	LPF(n+1)	Elkies/24
P-256	876	1	0.530	126.4	242	232	10
P-256	**1842**	1	0.723	126.9	92	215	13
P-224	777	1	0.607	110.5	145	182	14
P-224	1485	1	0.495	110.3	74	102	13
P-224	5736	1	0.540	110.4	204	142	11
P-224	**7026**	3	0.359	108.1	196	177	8
192	**139**	1	0.411	94.0	92	108	9
192	2198	1	0.498	93.6	176	149	11
192	**5228**	1	0.352	93.8	157	161	6
192	5405	1	0.434	94.0	132	177	10
192	**6013**	1	0.529	93.6	64	145	13
239	36	1	0.436	117.1	105	153	12
239	1562	1	0.590	118.1	219	136	17
239	**2185**	1	0.246	116.8	69	136	8
239	**2320**	5	0.476	115.2	128	123	10
239	2491	1	0.415	117.6	76	131	11
239	2682	5	0.558	115.7	170	227	17
239	2863	1	0.505	117.7	183	72	13
239	**3048**	1	0.410	117.5	110	151	12

Published curves in bold.

A.2 Legendre symbols and residues

run	cand	(n/p)	(p/n)	(t/p)	(t/n)	(b/p)	$(-3/n)$	$(-1/n)$	$(2/n)$	$n \,\%\, 8$	$p \,\%\, 8$	$t \,\%\, 4$
P-256	876	+	−	+	−	+	+	−	+	7	7	1
P-256	**1842**	−	−	+	−	+	+	+	+	1	7	3
P-224	777	+	+	−	+	+	+	−	−	3	1	3
P-224	1485	−	−	+	−	−	+	+	+	1	1	1
P-224	5736	−	−	+	−	−	+	+	+	1	1	1
P-224	**7026**	−	−	+	−	−	+	+	−	5	1	1
192	**139**	+	+	+	+	+	−	+	+	1	7	3
192	2198	+	+	+	+	+	+	+	+	1	7	3
192	**5228**	−	+	+	−	+	+	−	−	3	7	1
192	5405	+	+	−	+	+	+	+	+	1	7	3
192	**6013**	+	+	+	+	+	+	+	+	1	7	3
239	36	+	−	+	−	+	+	−	−	3	7	1
239	1562	−	+	−	+	+	+	−	+	7	7	1
239	**2185**	−	+	−	+	+	+	−	−	3	7	1
239	**2320**	−	+	+	+	+	+	−	−	3	7	1
239	2491	+	+	−	+	+	+	+	−	5	7	3
239	2682	+	−	+	−	+	−	−	+	7	7	1
239	2863	+	−	+	−	+	−	−	+	7	7	1
239	**3048**	+	+	+	+	+	−	+	+	1	7	3

A.3 Simulated early-abort rules ($\ell \le 100$)

rule	skipped rejected	published rejected
(a) $n \equiv 0$ — correct abort (control)	0/11	0/8
(b) $p+1+t \equiv 0$ — trace sign flipped	9/11	7/8
(c) order of the other root of b	6/11	6/8
(d) sign and root both flipped	3/11	1/8
(e) $t \equiv 0$	5/11	4/8
(f) $t \equiv \pm 1, \pm 2$	11/11	8/8
(g1) $n \equiv 1$ (tests $n-1$)	11/11	8/8
(g2) $n \equiv -1$ (tests $n+1$)	10/11	7/8
(g3) $\lvert t \rvert \equiv 0$	5/11	4/8
(h) twist order $\equiv 0$	9/11	7/8

A.4 Small multiples of the published generators

Shortest coordinate among kG ($k \le 2^{16}$) and G/k ($k \le 2^{16}$ prime-field, $\le 2^{12}$ binary), negations included; "expected" is the minimum of that many uniform field elements.

curve	h	$x(G/h)$ bits	shortest found (bits)	expected
P-192	1	–	174	175
prime192v2	1	–	175	175
prime192v3	1	–	176	175
P-224	1	–	208	207
prime239v1	1	–	222	222
prime239v2	1	–	220	222
prime239v3	1	–	221	222
P-256	1	–	237	239
P-384	1	–	367	367
P-521	1	–	502	504
B-163	2	163	147	147
K-163	2	**126**	**126** ($G/2$)	147
B-233	2	232	215	217
K-233	4	full	219	217
B-283	2	283	267	267
K-283	4	full	267	267
B-409	2	408	391	393
K-409	4	full	393	393
B-571	2	571	554	555
K-571	4	full	555	555

For the $h = 4$ Koblitz curves $G/4$ lies inside the G/k scan and no coordinate within 32 bits of the field size was found ("full").